

Multi-View Convolutional Neural Network for Musculoskeletal Disorder Detection in Radiographic Imaging

Camille Trouvé¹, Ceren Dinc¹, Esteban Boulanger¹ Roger El Achkar¹

¹*Faculté de Gestion Economie et Sciences - Ecole du Numérique*

January 11, 2026

Abstract

Musculoskeletal disorders are among the main causes of disability, according to the World Health Organization. This study aims to develop a tool capable of determining abnormalities in musculoskeletal radiographs. The proposed model is a convolutional neural network based on ResNet18 architecture with a multiple-view fusion module to integrate multiple radiographic projections, mimicking how radiologists interpret radiographic views. The model was trained and tested on the MURA dataset (Stanford), a large open-access dataset combining 40,561 radiographic images of the upper extremity from multiple studies. Through systematic evaluation of seven experimental configurations, the best-performing model achieved a validation AUC-ROC of 0.6545, representing a substantial improvement of +0.0906 compared to the baseline.

1 Introduction

Musculoskeletal conditions represent a profound global health challenge, affecting approximately 1.7 billion people and serving as the primary driver of disability worldwide[1]. These disorders, which include fractures, osteoarthritis, and various degenerative joint diseases, significantly impair mobility and dexterity, often leading to premature exit from the workforce and a diminished quality of life. The increasing prevalence of these conditions, exacerbated by a growing and aging population, has placed an unprecedented strain on healthcare systems[2]. Radiologists workloads rise because of a higher volume of complex images, which contributes to fatigue and a statistically significant risk of diagnostic error[1, 3, 4].

In this context, artificial intelligence (AI), specifically deep learning (DL) utilizing convolutional neural networks (CNNs), has emerged as a tool for diagnostic assistance[2, 3, 5]. CNN architectures are uniquely suited for image-based problems because they can autonomously learn discriminative features from large datasets, matching or even exceeding human performance in narrow tasks[3, 4]. Within musculoskeletal radiology, DL applications have expanded across four primary categories: lesion detection, classification, segmentation, and non-interpretive tasks such as protocol optimization. While expert-level performance has been achieved in specific domains like bone age assessment

and certain fracture detections, comprehensive interpretation of complex imaging examinations remains an ongoing challenge[3, 5].

A pivotal milestone in this field was the release of the MURA (Musculoskeletal Radiographs) dataset, a large-scale collection of 40,561 upper extremity images manually labeled by board-certified radiologists[1]. Initial benchmark studies using a 169-layer DenseNet achieved high overall AUC-ROC scores but revealed a significant performance gap in specific anatomical regions, particularly the finger and humerus radiographs, where the model struggled to match the agreement levels of practicing radiologists. Furthermore, many existing models operate on single radiographic projections, whereas clinical diagnosis typically necessitates the integration of multiple views to resolve spatial ambiguities[1, 5].

This study proposes a multi-view neural network architecture designed to bridge this gap by mimicking the diagnostic process of human experts. By employing a ResNet18 backbone pretrained on ImageNet[6] and a specialized masked average pooling module, the model integrates features from multiple radiographic views into a singular study-level prediction.

2 Material and Method

2.1 Dataset

In this project, we used the MURA dataset (Musculoskeletal Radiographs), which contains X-ray images of upper limbs labeled as either normal or abnormal[1]. Labels are provided at the study level, meaning that all images belonging to the same patient examination share the same label.

Each study S_i contains between one and four radiographic views:

$$S_i = \{x_i^{(v)}\}_{v=1}^{n_i}, \quad n_i \in \{1, 2, 3, 4\}, \quad (1)$$

where $x_i^{(v)}$ denotes the v -th image of study i . Each study is associated with a binary label $y_i \in \{0, 1\}$ indicating whether it is abnormal.

Although the full dataset contains more than 40,000 images, we restricted training to a subset of 1,000 studies to reduce computational cost. The official validation split was used for evaluation.

2.2 Preprocessing and Data Augmentation

All images were originally stored as grayscale images and were converted to three channels to match the input format of pretrained convolutional neural networks. Images were resized to 224×224 pixels.

Pixel intensities were normalized using a fixed mean and standard deviation of (0.5, 0.5, 0.5). During training, simple data augmentation techniques such as random horizontal flips were applied.

Since the number of views per study varies, we padded each study to a maximum of four images and used a binary mask:

$$m_i^{(v)} = \begin{cases} 1 & \text{if view } v \text{ exists} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

to indicate valid views.

2.3 Model Architecture

We implemented a multi-view classification model in which each image is processed independently and then aggregated at the study level.

2.3.1 Feature Extraction

Each image is passed through a shared ResNet18 network pretrained on ImageNet[6]. The final classification layer is removed, and the output of the network is a feature vector:

$$h_i^{(v)} = f_\theta(x_i^{(v)}) \in \mathbb{R}^{512}. \quad (3)$$

2.3.2 Multi-View Aggregation

The features of all views belonging to the same study are combined using a masked average pooling operation:

$$z_i = \frac{1}{\sum_v m_i^{(v)}} \sum_{v=1}^4 m_i^{(v)} h_i^{(v)}. \quad (4)$$

This produces a single feature vector z_i representing the entire study.

2.3.3 Classification

The aggregated feature vector is then passed through a two-layer feedforward classifier with ReLU activation and dropout for regularization, followed by a sigmoid activation to obtain the predicted probability of abnormality:

$$\hat{y}_i = \sigma(W_2 \text{ReLU}(W_1 z_i + b_1) + b_2), \quad (5)$$

where $W_1 \in \mathbb{R}^{256 \times 512}$, $W_2 \in \mathbb{R}^{1 \times 256}$, and dropout with rate 0.3 is applied after the ReLU activation. The final output $\hat{y}_i \in (0, 1)$ represents the predicted probability of abnormality.

2.4 Training Procedure

To handle class imbalance, we used a weighted binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B [w_+ y_i \log(\hat{y}_i) + w_- (1 - y_i) \log(1 - \hat{y}_i)]. \quad (6)$$

The model was trained using the Adam optimizer[7] with a batch size of 4. Learning rates were varied across experiments (3×10^{-4} , 1×10^{-3} , and 3×10^{-3}) to investigate their impact on model performance.

2.5 Evaluation Metrics

Model performance was mainly evaluated using the Area Under the Curve Receiver Operating Characteristic (AUC-ROC). In addition, accuracy, sensitivity, and specificity were

computed. For a given decision threshold τ , these metrics are defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \quad (7)$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (8)$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \quad (9)$$

3 Results

3.1 Experimental setup and evaluation metric

The proposed multi-view model was evaluated on the official validation split of the MURA dataset using study-level labels (normal vs. abnormal). Given the class imbalance commonly encountered in medical imaging, performance was primarily assessed using the validation AUC-ROC, as shown in Figure 1 and Figure 2. To ensure fair comparison across configurations, all experiments were trained using 10 epochs.

3.2 Overall comparison of experimental configurations

Seven experimental configurations were compared by varying the learning rate, batch size, and number of training samples. Performance was summarized using the best validation AUC-ROC achieved during training. The configuration with a reduced learning rate (*low_lr*) achieved the highest performance, with an AUC-ROC of 0.6545, corresponding to an absolute improvement of +0.0906 compared to the baseline configuration (AUC-ROC = 0.5639). Intermediate improvements were observed for the *low_data* (AUC-ROC = 0.6076) and *large_batch* (AUC-ROC = 0.6039) configurations, whereas the *more_data* configuration underperformed relative to the baseline (AUC-ROC = 0.5409).

3.3 Effect of the learning rate

At a fixed training set size of 1000 samples and a constant training budget of 10 epochs, reducing the learning rate from 1×10^{-3} to 3×10^{-4} resulted in the most substantial improvement in validation AUC-ROC (0.5639 to 0.6545). Increasing the learning rate to 3×10^{-3} led to only a limited improvement in performance. Among the hyperparameters explored, the learning rate therefore appears to have the strongest influence on validation performance.

3.4 Effect of batch size

When varying the batch size while keeping the learning rate fixed at 1×10^{-3} and using 1000 training samples, the configuration with a larger batch size achieved a higher validation AUC-ROC (0.6039) than the small-batch configuration (0.5679). This suggests a moderate benefit from larger batch sizes in this experimental setting.

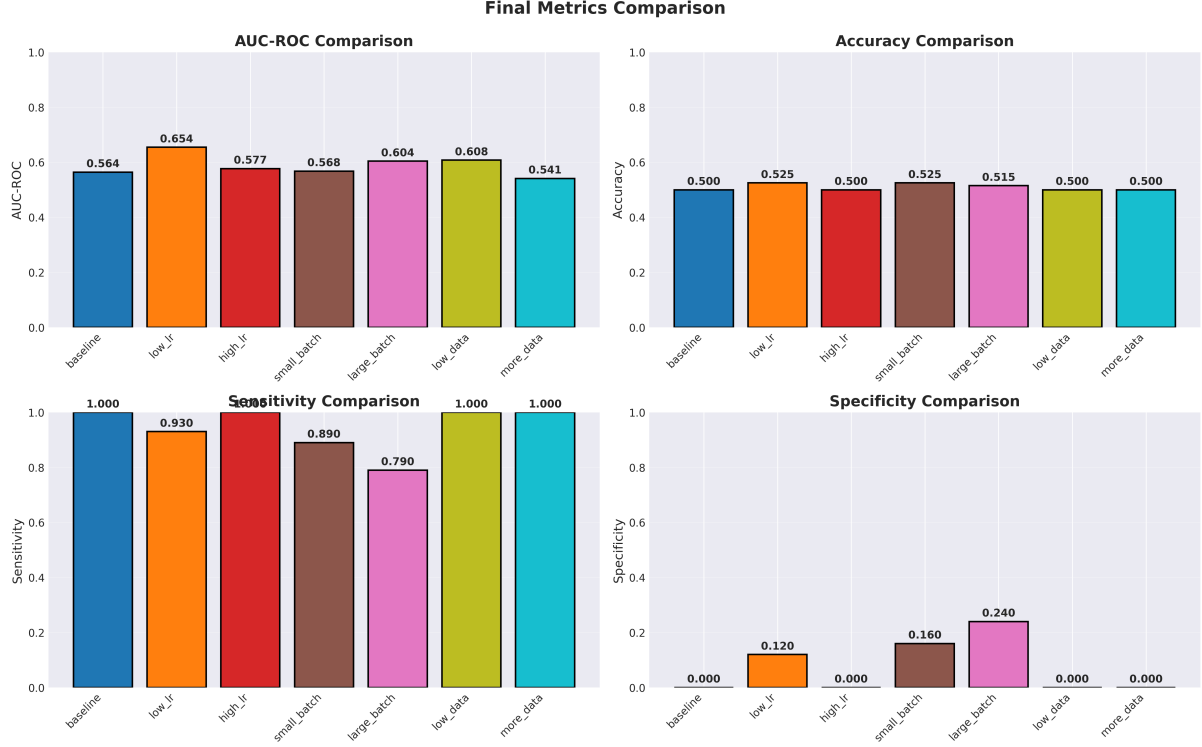


Figure 1: Comparison of final performance metrics across all experimental configurations. The figure shows AUC-ROC and accuracy for each configuration.

3.5 Effect of training dataset size with fixed epochs

Varying the size of the training dataset (500, 1000, and 2000 samples) while keeping the number of epochs fixed did not result in a consistent improvement in validation AUC-ROC. The *low_data* configuration, trained on 500 samples, achieved an AUC-ROC of 0.6076, whereas increasing the training set size to 2000 samples (*more_data*) resulted in a lower AUC-ROC of 0.5409. Under this fixed-epoch protocol, increasing the number of training samples did not translate into improved ranking performance.

3.6 Training efficiency and threshold-based metrics

In terms of computational efficiency, the *low_data* configuration had the shortest training time (23.6 minutes) and achieved the highest AUC-ROC per minute, making it suitable for rapid experimentation. At the selected decision threshold, the best-performing configuration (*low_lr*) achieved an accuracy of 0.5250, with a sensitivity of 0.93 and a specificity of 0.12, indicating a strongly recall-oriented operating point.

4 Discussion

This study focused on the development and evaluation of a multi-view convolutional neural network based on a ResNet18 backbone for automated abnormality detection in musculoskeletal radiographs. The main objective was to explore whether integrating multiple radiographic projections at the study level could improve the classification of upper extremity X-ray examinations as normal or abnormal. Such an approach is motivated by

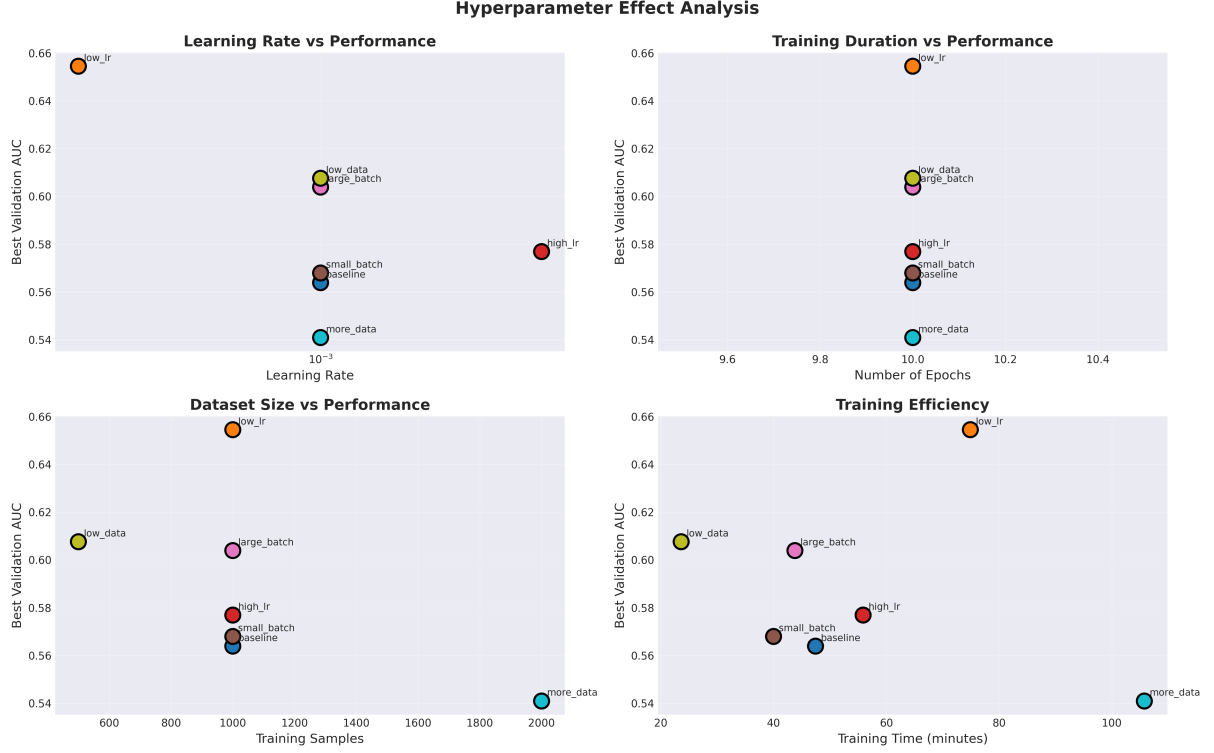


Figure 2: Hyperparameter effect analysis across experimental configurations. The four panels show the relationship between (top-left) learning rate and validation AUC-ROC, (top-right) training duration and performance, (bottom-left) dataset size and performance, and (bottom-right) training efficiency measured as AUC-ROC per training time.

clinical practice, where radiologists rely on complementary views when interpreting musculoskeletal imaging, and aims to investigate the potential of deep learning as a diagnostic support tool.

The best-performing configuration reached a validation AUC-ROC of 0.6545, indicating a moderate ability to discriminate between normal and abnormal studies. Although this level of performance is not sufficient for direct clinical deployment, it represents a reasonable proof of concept given the constraints of this work. In particular, the model was trained using a limited computational budget and a relatively lightweight architecture. The results suggest that the network was able to combine information from multiple views through the proposed pooling strategy, which is consistent with the intended multi-view design.

From a clinical perspective, the operating characteristics of the model highlight important trade-offs. The configuration with the highest AUC-ROC achieved a high sensitivity of 0.93, meaning that most abnormal cases were correctly identified. This property could be valuable in screening-oriented settings, where minimizing missed abnormalities is a priority. However, this high sensitivity was associated with a very low specificity (0.12), leading to a large number of false positives. As a result, the current model would not be suitable as a standalone diagnostic system.

The observed performance is likely influenced by several methodological and computational limitations. Training was restricted to a subset of 1,000 studies, a fixed budget of 10 epochs, and a ResNet18 architecture in order to accommodate CPU-based training. These constraints may have limited the model’s capacity to fully learn complex

and diverse pathological patterns present in the dataset. Despite these limitations, the experiments showed that optimization choices, particularly the learning rate, had a noticeable impact on performance, suggesting that more extensive hyperparameter tuning could lead to further improvements.

Several limitations of this study should be noted. The model was trained and evaluated on a limited subset of the data, which may reduce its generalization to unseen cases. Each configuration was evaluated using a single training run, preventing an assessment of performance variability across different random initializations. In addition, evaluation was restricted to the validation set, and no independent test set was used. Finally, the current approach does not provide interpretability mechanisms to understand which image regions or anatomical structures contribute to the model’s predictions, which is an important aspect for clinical acceptance.

Future work should aim to extend this study by leveraging greater computational resources. This would allow training on the full MURA dataset, increasing the number of training epochs, and exploring more expressive architectures such as deeper convolutional networks or transformer-based models. Incorporating learning rate scheduling, data augmentation, and systematic hyperparameter optimization could also improve performance. Furthermore, adding interpretability methods such as Grad-CAM would help assess whether the model focuses on clinically relevant regions across views. More robust evaluation protocols, including multi-seed experiments, cross-validation, and testing on an independent dataset, would be necessary to better assess generalization and reliability. Ultimately, these extensions would help clarify the practical relevance of multi-view deep learning approaches for musculoskeletal abnormality detection.

References

- [1] Pranav Rajpurkar et al. *MURA: Large Dataset for Abnormality Detection in Musculoskeletal Radiographs*. May 22, 2018. DOI: 10.48550/arXiv.1712.06957. arXiv: 1712.06957[physics]. URL: <http://arxiv.org/abs/1712.06957> (visited on 10/30/2025).
- [2] Govind Chada. “Machine Learning Models for Abnormality Detection in Musculoskeletal Radiographs”. In: *Reports* 2.4 (Dec. 2019). Publisher: Multidisciplinary Digital Publishing Institute, p. 26. ISSN: 2571-841X. DOI: 10.3390/reports2040026. URL: <https://www.mdpi.com/2571-841X/2/4/26> (visited on 10/30/2025).
- [3] Pauley Chea and Jacob C. Mandell. “Current applications and future directions of deep learning in musculoskeletal radiology”. In: *Skeletal Radiology* 49.2 (Feb. 1, 2020), pp. 183–197. ISSN: 1432-2161. DOI: 10.1007/s00256-019-03284-z. URL: <https://doi.org/10.1007/s00256-019-03284-z> (visited on 10/30/2025).
- [4] Gurpreet Singh et al. “Hybrid Deep Learning Approach for Automatic Detection in Musculoskeletal Radiographs”. In: *Biology* 11.5 (May 2022). Publisher: Multidisciplinary Digital Publishing Institute, p. 665. ISSN: 2079-7737. DOI: 10.3390/biology11050665. URL: <https://www.mdpi.com/2079-7737/11/5/665> (visited on 10/30/2025).

- [5] Tommaso D'Angelo et al. "Artificial intelligence, machine learning and deep learning in musculoskeletal imaging: Current applications". In: *Journal of Clinical Ultrasound* 50.9 (2022). _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/jcu.23321>, pp. 1414–1431. ISSN: 1097-0096. DOI: 10.1002/jcu.23321. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/jcu.23321> (visited on 10/30/2025).
- [6] Kaiming He et al. "Deep Residual Learning for Image Recognition". In: *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). ISSN: 1063-6919. June 2016, pp. 770–778. DOI: 10.1109/CVPR.2016.90. URL: <https://ieeexplore.ieee.org/document/7780459> (visited on 01/11/2026).
- [7] Diederik P. Kingma and Jimmy Ba. *Adam: A Method for Stochastic Optimization*. Jan. 30, 2017. DOI: 10.48550/arXiv.1412.6980. arXiv: 1412.6980[cs]. URL: <http://arxiv.org/abs/1412.6980> (visited on 01/11/2026).